

C-POSE

Geometry Coregistration for Mars 2020

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18 December 2019



C

Camera

P

Positions

O

(and) Orientations

S

(in) Surface

E

Environments



- Geometry coregistration server
- API for storage, retrieval, ingestion, and search of pointing models
- Search across different cameras
- Different “opinions” about what a given pointing model’s values should be
 - For a given camera
 - At a certain RMC
- “Opinions” are registered as **Solutions**
- A set of Solutions across cameras and RMCs is called a **View**
- A combination of an RMC and a camera is called a **Pose**



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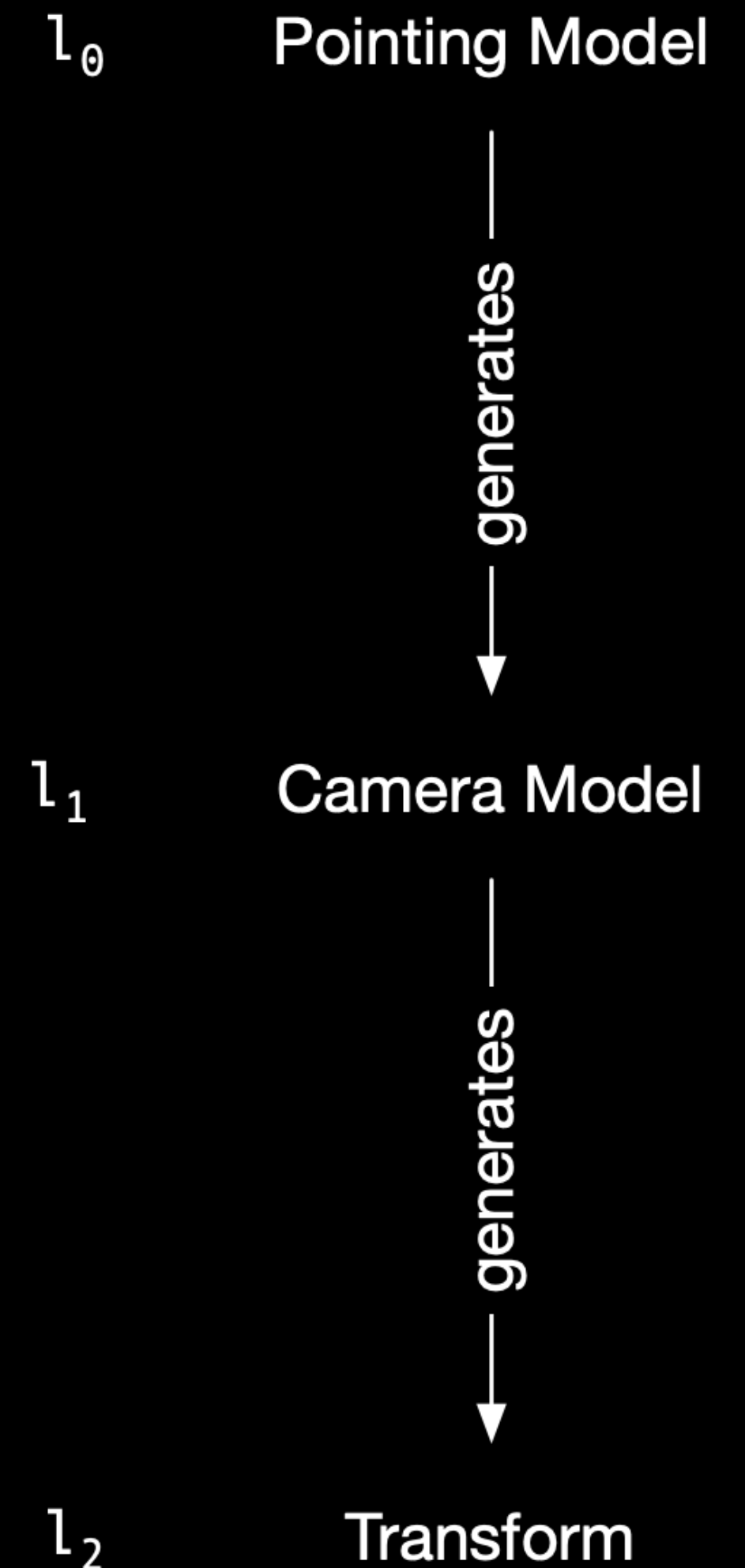
(in) Surface

E

Environments



- A **Solution** consists of three “levels” of data:
 1. Pointing Models
 2. Camera Models
 3. Transforms
- Given #1, both #2 and #3 will be generated and stored
- Given #2, only #3 will be generated, and #1 will be null
- Given #3, both #1 and #2 will be null
- If any “level” of a Solution is updated with new information, levels “below” it will be updated



- Multiple **Solutions** may exist for a given **Pose**, even within a given **View**
- View states differentiated by timestamps
- Enables queries such as:

“As of 12:03 P.M. on December 13th, 2019, what was the state of my View?”

- Analogous to solution IDs in PLACES

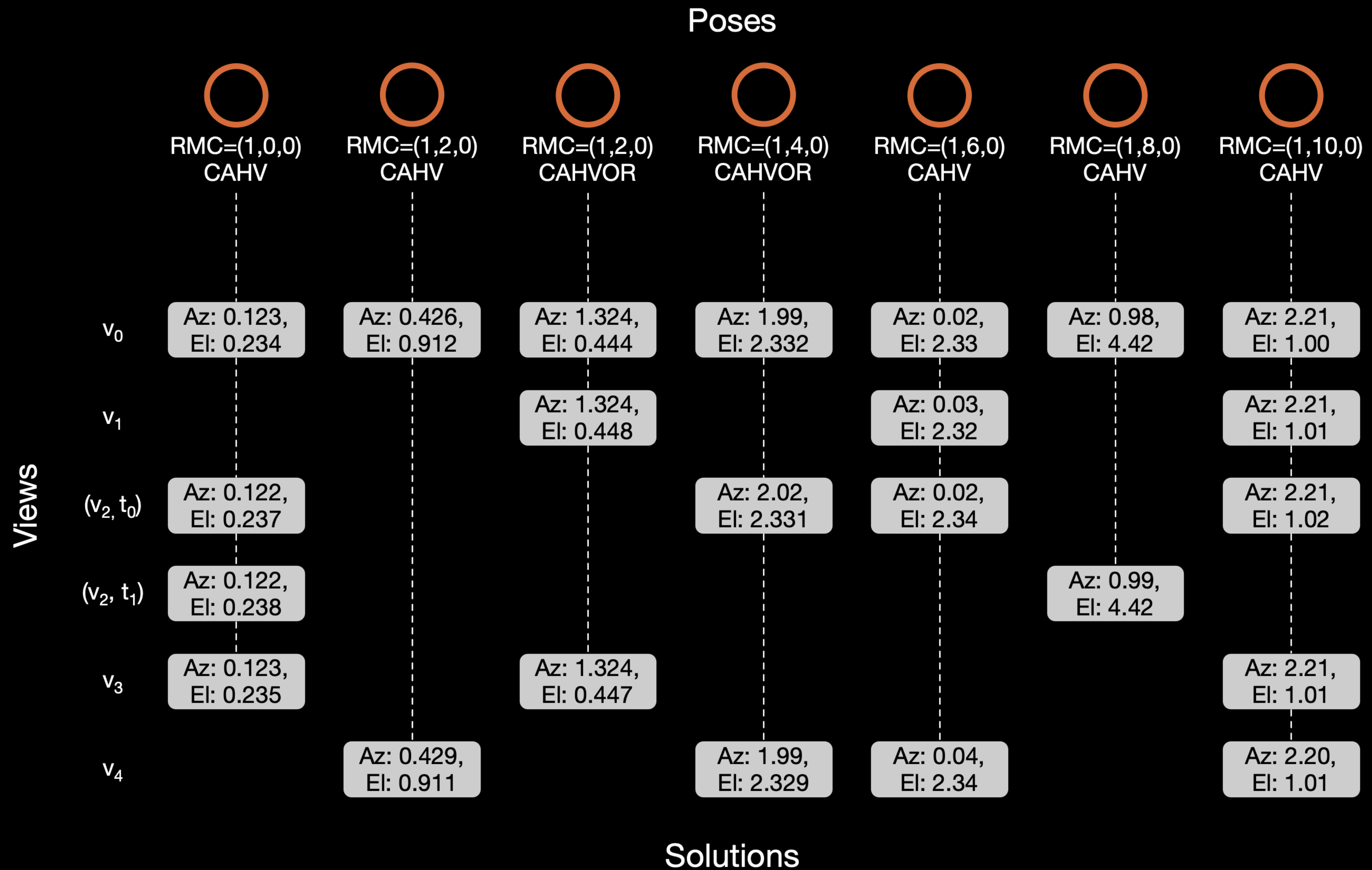


- **Views** may inherit **Solutions** from other **Views** if they do not have a one for a given **Pose**
- Enables scenarios such as:

I agree with Bob's opinion, except for the pointing model at $RMC=(1,40,0)$.

- A new View may be created that is identical to Bob's View except for your new Solution
- Analogous to View inheritance in PLACES





C-POSE Homepage



Welcome to C-POSE (EN)

<https://github.jpl.nasa.gov/MIPL/C-POSE> [Documentation](#)

Defined Routes

METHOD	PATH	NAME	HANDLER
GET	/	rootPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.HomeHandler
GET	/api/	apiPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.ApiServe
POST	/images/ingest/	imagesIngestPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.ImagesIngest
POST	/poses/	posesPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Create
GET	/poses/	posesPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.List
GET	/poses/new/	newPosesPath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.New
GET	/poses/{instrument_pose_id}/	posePath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Show
DELETE	/poses/{instrument_pose_id}/	posePath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Destroy
PUT	/poses/{instrument_pose_id}/	posePath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Update
GET	/poses/{instrument_pose_id}/edit/	editPosePath	github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Edit
			github.jpl.nasa.gov/MIPL/C-POSE/actions.InstrumentPosesResource.Edit



C-POSE API

Search...

Authentication

POST

Submit a new image ingest job to C-POSE

GET

Retrieve Solutions

POST

Submit Solution(s) to C-POSE

GET

Retrieve a View by its name

GET

Retrieve a list of all Views

POST

Create a new View

Documentation Powered by ReDoc

Submit Solution(s) to C-POSE

Submit Solution(s) to C-POSE

REQUEST BODY SCHEMA: application/json

Array [

instrument_pose_id

string

>= 0 characters

transform

>object (transform)

pointing_model

<object (pointing_model)

values

Array of numbers <float>

camera_model

>object (camera_model)

Responses

201

Successfully created Solution(s)

Retrieve a View by its name

Retrieve a View by its name

Responses

POST

/solutions

Request samples

Payload

Content type
application/json

CopyExpand allCollapse all

```
{  "instrument_pose_id": "st",  "transform": {    "x": 0,    "y": 0,    "z": 0  },  "pointing_model": {    "x": 0,    "y": 0,    "z": 0  },  "camera_model": {    "x": 0,    "y": 0,    "z": 0  }}
```

GET

/views/{name}

Response samples

200



C-POSE GoDocs

localhost:6060/pkg/github.jpl.nasa.gov/MIPL/C-POSE/utils/#IngestSolutionsFromS3URIs

The Go Programming Language

DocumentsPackagesThe ProjectHelpBlogSearch

Package utils

import "github.jpl.nasa.gov/MIPL/C-POSE/utils"

OverviewIndex

Overview

Index

func CSSOMiddleware(next buffalo.Handler) buffalo.Handler

func DownloadFileFromS3(uri string, sess client.ConfigProvider) (error, string)

func IngestSolutionFromS3URI(msg sqs.Message, c chan int32, jobID uuid.UUID, sess client.ConfigProvider)

func IngestSolutionsFromS3URIs(output sqs.ReceiveMessageOutput, jobID uuid.UUID, sess client.ConfigProvider) error

func ParseIngestJobFromSQS() error

func SendIngestJobToSQS(imagesList []string, toView string, c chan int32, ctx buffalo.Context)

func SendIngestJobsToSQS(imagesList []string, toView string, ctx buffalo.Context) error

type CSSOHeaders

func (csso CSSOHeaders) String() string

Package files

csso.go ingest.go s3.go sqs.go

func CSSOMiddleware

func CSSOMiddleware(next buffalo.Handler) buffalo.Handler

func DownloadFileFromS3

func DownloadFileFromS3(uri string, sess client.ConfigProvider) (error, string)

func IngestSolutionFromS3URI

